DATASCI 3ML3 Final Project

Wikipedia Article Classification

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1 Overview and Motivation

On Wikipedia, before articles are made visible to the public, they must go through a review process. Reviewers, who are volunteers, must follow a 30+ step flowchart for each published article;[1] this time-consuming process, along with a shortage of reviewers, means there is often a months-long waiting time for articles to be reviewed (at the time of writing, the number of articles waiting to be reviewed is over 17,000). Therefore, any additional tools that could shorten the time taken to review an article would be incredibly beneficial to the Wikipedia community.

For this project, we will create a model which can handle a step of this process. Articles can be classified into one or more *WikiProjects*; a WikiProject is a group of editors focusing on a specific topic. For example, WikiProject Canada is a group of (mainly Canadian) editors improving Canada-related articles. Classifying articles into projects is extremely important, since it allows them to be targeted for improvement by editors knowledgeable about the subject area; unclassified articles are often neglected. However, since classifying is considered an "optional" step in the reviewing process, many articles end up unclassified.

To this end, we will create a model to solve this multi-label classification problem. This is different from multi-class classification, since an article can be part of zero, one, or more WikiProjects. Since there are thousands of projects of varying size (for example, *Computing* has 36,000 articles; *Artificial intelligence* has 1,600; and *Human-computer interaction* has only 184), we will focus on classifying only the approximately 100 largest projects as a proof-of-concept. However, we will explore next steps for a full model in the conclusion.

We will use Tensorflow/Keras to build and train the model, along with `scikit-learn` for its natural language processing features (specifically, TF-IDF). In the remainder of this report, we will discuss data acquisition and processing, creating a model, evaluation and experimentation; finally, we will create an improved model based on these insights.

2 Data Acquisition and Preprocessing

To train our model, we need data in the format $(lead, projects)$, where the *lead* is the beginning article text, and *projects* is a list containing the WikiProjects that article has been assigned. While there is no existing dataset containing the exact data we need for this model, Wikipedia is very open and provides a number of APIs which we can use to collect the data ourselves. We will collect only the lead paragraphs of an article, instead of its full content, as its projects can be determined from the first paragraph alone in the vast majority of cases.

Wikipedia provides a directory of all WikiProjects, along with details about their size.[2] To simplify the model, we will classify only the 25 WikiProjects with the largest number of articles. Our first goal is to find a list of articles in each project; we can use the WP 1.0 API for this.

A major problem that we have to deal with here is the relative sizes of each category. WikiProject Biography, for example, has nearly 2.2 million articles; while a smaller WikiProject Lepidoptera has only around 100,000. This difference would be even more pronounced if we were to classify more than 25 projects: the vast majority of projects have fewer than 10,000 articles. If we were to use all possible data to train our model, it would likely have a bias towards larger projects while potentially ignoring smaller ones.[3]

To combat this, we use a technique called undersampling. While there are fairly complex algorithms that can be used for this purpose, we will use the simpler *random undersampling*, which will result in a more balanced dataset. The way this works in practice is by training on only a portion of the available dataset for majority classes. So, we will obtain only a random subset of articles in each project: approximately 2,000 from each. This is likely far more than needed to train an effective model; in the Evaluation section below, we will explore training models with varying amounts of available data.

After finding a list of the articles from each project, we will use Wikipedia's `extracts` API to fetch the lead section for each. Finally, since an article can be part of more than one project, we go through each article and determine its other projects. Unfortunately, Wikipedia does not provide a convenient API for

[1] See `https://upload.wikimedia.org/wikipedia/commons/f/f4/NPP_flowchart.svg`.

[2] See `https://en.wikipedia.org/wiki/Wikipedia:WikiProject_Directory/All`.

[3] O. Olawale Awe, "Computational Strategies for Handling Imbalanced Data in Machine Learning"

this purpose; so instead, we use regular expressions to parse the markup code for each article. We then save all of our collected data to `all_data.csv`. An example row in this dataset is the following:

```
Treaty of London (1867),"The Treaty of London (French: Traite de Londres), often
called the Second Treaty of London after the 1839 Treaty, granted Luxembourg full
independence...",WikiProject London|WikiProject Luxembourg|...
```

3 Feature Engineering

At the core of our model is the TF-IDF (term frequency-inverse document frequency) algorithm. Unlike the simpler bag-of-words model, which only captures the frequency of a term over a dataset, TF-IDF measures the relative importance of each term. We define $f_{t,d}$ as the number of times term t occurs in document d; we then define the *term frequency* as

$$\mathrm{TF}(t,d) = \frac{f_{t,d}}{\sum_{t' \in d} f_{t',d}}$$

Or, in simpler terms, TF represents the fraction of words that are t in d. The other metric we need to consider is *inverse document frequency*. Let D be the set of all documents in our dataset:

$$\mathrm{IDF}(t,D) = \log \frac{|D|}{|\{d \in D : t \in d\}|}$$

This measures the relative rarity of words across the dataset: higher values means the term occurs in fewer documents. A logarithm is used because without it, a word that appears once in a million documents would have an IDF score of 1,000,000, resulting in the algorithm weighing it much higher than it should. We then multiply the TF and IDF values together to get the final result:

$$\text{TF-IDF}(t,d,D) = \mathrm{TF}(t,d) \times \mathrm{IDF}(t,D)$$

For example, consider the word "the". As one of the most commonly used words in English, it appears in approximately 96% of the leads in our dataset; this would result in its IDF score being $\log \frac{100}{96} \approx 0.02$, or an extremely low "importance". However, if we consider the word "Canadian", which only appears in 4% of leads, its IDF would be $\log \frac{100}{4} \approx 1.4$.

For our model, we will use the `scikit-learn` library's `TfidfVectorizer` class. This has a number of parameters to improve the algorithm's accuracy and performance:

- `max_features` limits the number of terms, selecting only the top N features by their term frequency. Setting this limit reduces the dimensionality of the data, which can reduce memory usage and prevent overfitting by excluding rare words.
- `stop_words` excludes common English words which do not provide a significant amount of information, helping the model focus on more important terms. This can also have the benefit of reducing the number of features and thus improving performance.
- `ngram_range` is a tuple (min, max) that tells the vectorizer how many tokens in a row to consider as a single term. For example, with the input "New York City", the range $(1,1)$ would produce `["New", "York", "City"]`, whereas $(1,3)$ would also consider `"New York"` and `"New York City"`, allowing for more semantic context. However, this comes at the cost of a significantly higher number of features.

Importantly, when fitting the vectorizer to the data, we use only the training data, and not any test data, to ensure there is no data leakage.

4 Model Implementation

5 Evaluation and Insights

5.1 Experiment: Data Volume

Since our model is classifying no more than the 100 largest WikiProjects, all of which have at least 20,000 articles, our model is not suffering from a lack of training data. However, should the model be expanded in the future to much smaller projects, some of which have only hundreds of examples, this could become a problem. Therefore it would be beneficial to determine how much data is needed to make accurate predictions.

Figure 1 shows the results of the experiment. The chart shows the classification metrics (Precision, Recall, and F1-Score) graphed against the portion of the dataset used for training: between 1% and 90%. To ensure a fair comparison between these radically different dataset sizes, instead of artificially constraining the number of epochs, the Keras callback `EarlyStopping` was used to stop training when the validation accuracy decreased. This ensures that models with less data do not overfit due to their limited sample size.

Figure 1: Precision, Recall, and F1-score evaluated across different data splits.

While the model's performance when using less than 10% of the dataset is extremely poor, the accuracy begins to level out around 20%, and by 50% we experience diminishing returns and any improvement becomes very marginal. While increasing the amount of data used does result in an improvement, even if small, it comes at a cost: the training time increases significantly when more data is used. For example, when all available data was used to train a model, the training time was roughly 3x that of a model using only 30%; however, the improvement in accuracy was only 2%. Thus it may not be worth it to always use the largest possible dataset.

5.2 Experiment: Dropout

Dropout is a regularization technique used in neural networks to prevent overfitting. It entails deactivating a random portion of neurons when training, forcing the network to learn robust patterns instead of memorizing the training data. In this experiment, we train models with different levels of dropout and examine their training and validation loss throughout the process. This experiment uses models that have low, medium, and high levels of dropout (0.0, 0.3, and 0.6 respectively).

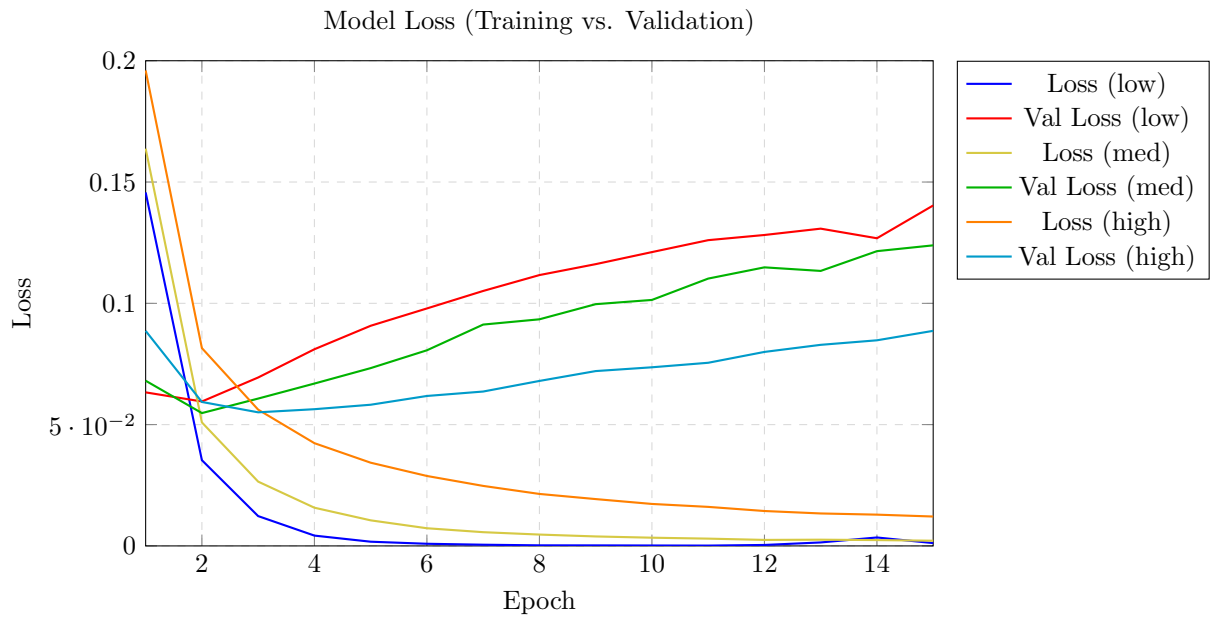

Figure 2: Training and validation loss across epochs for different dropout rates.

5.3 Experiment: Max Features

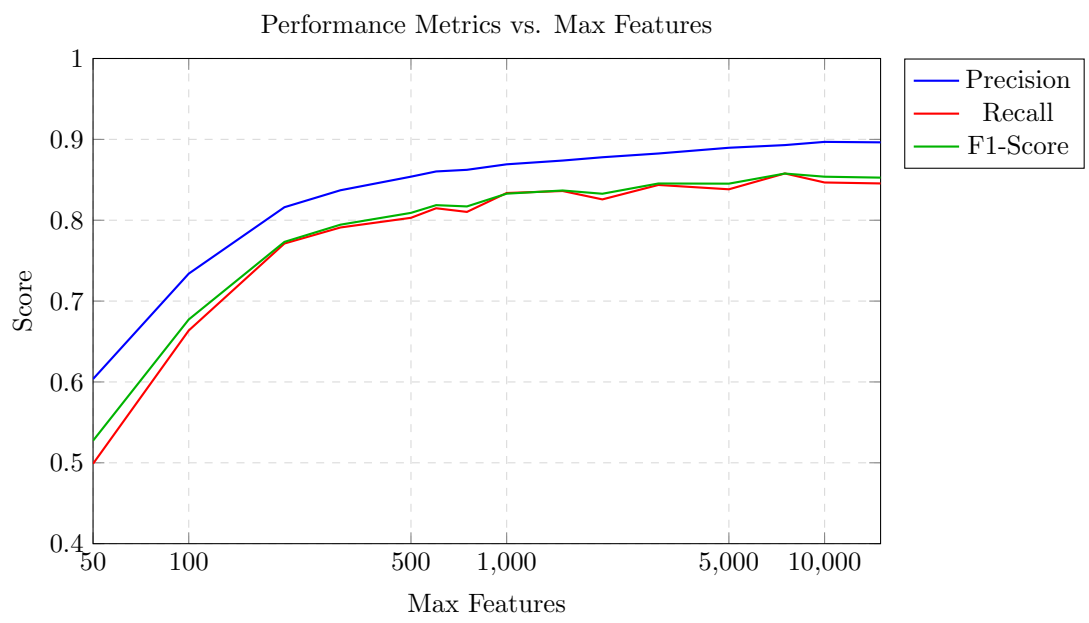

Figure 3: Performance metrics evaluated across an increasing count of max features (logarithmic scale).

6 Conclusion

6.1 Next Steps

- Partial random under-sampling[4]

[4]Nicolás García-Pedrajas, "Partial random under/oversampling for multilabel problems"

- Pretrained models or word embeddings
- Categories instead of projects

7 Code Organization

The code for this project is split between Python modules and Jupyter notebooks: testing, experiment, and data collection code is in notebooks, while the modularized `Model` class is in a Python file. The description for each file is as follows:

- `data_collection.ipynb` uses the steps described in Section 2 to collect data: it uses Wikipedia's API to collect articles in each project, find the lead text for each, find other WikiProjects for each article, and then saves the result to `all_data.csv`.
- `create_model.ipynb` walks through the steps for creating a model: using a binarizer to turn WikiProjects into vectors, splitting test and training data, vectorizing the text, and creating, training, and saving the model.
- `model.py` generalizes the code from this notebook to make a modular interface with a `Model` class. This allows us to pass in additional parameters to run experiments: dataset size, training duration, dropout, etc. It also provides `save`, `load`, and `get_report` functions, so we can easily tell how a model is performing, or load a previously trained model.
- `evaluation.ipynb` contains a function, `run_eval_thread`, which uses the Model class to create and train a model with the given parameters. It does this on a separate thread, to ensure memory is reclaimed once it is done. It then runs the model's `get_report` function, and returns statistics.

8 References